

Chapter 25 & 26: Diagonalization and Similarity

Definition: If A and B are $n \times n$ matrices, we say A and B are **similar**, and write $A \sim B$ if there is an invertible matrix P with $B = P^{-1}AP$.

Fact: If $A \sim B$, then $B \sim A$.

Proof: $A \sim B \Rightarrow B = P^{-1}AP \Rightarrow A = PBP^{-1} \Rightarrow B \sim A$.

Properties shared by similar matrices

Definition: The **trace** $\text{tr}(A)$ of an $n \times n$ matrix A is the sum of the entries on the main diagonal.

Example: The trace of $A = \begin{bmatrix} -2 & 2 & -4 \\ -7 & 3 & 6 \\ 4 & 1 & 0 \end{bmatrix}$ is $\text{tr}(A) = -2 + 3 + 0 = 1$.

Fact: For two $n \times n$ matrices A and B and scalar c ,

- a. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.
- b. $\text{tr}(cA) = c \text{tr}(A)$
- c. $\text{tr}(AB) = \text{tr}(BA)$ (However, $\text{tr}(AB)$ does NOT necessarily equal $\text{tr}(A) \text{tr}(B)$).

Proof of (c) (if time): $\text{tr}(AB) = \sum_{i=1}^n a_{ii}b_{ii} + \sum_{i=1}^n a_{2i}b_{i2} + \dots + \sum_{i=1}^n a_{ni}b_{in} = \sum_{j=1}^n \sum_{i=1}^n a_{ji}b_{ij}$

$$\begin{aligned} \text{tr}(BA) &= \sum_{i=1}^n b_{ii}a_{ii} + \sum_{i=1}^n b_{2i}a_{i2} + \dots + \sum_{i=1}^n b_{ni}a_{in} \\ &= \sum_{j=1}^n \sum_{i=1}^n b_{ji}a_{ij} \Rightarrow \text{tr}(AB) = \text{tr}(BA) \end{aligned}$$

Theorem: If A and B are similar matrices, then A and B have the same determinant, rank, trace, characteristic polynomial, and eigenvalues.

Proof of Trace:

$$\text{tr}(B) = \text{tr}(P^{-1}AP) \stackrel{\substack{\uparrow \\ \text{by prev. fact}}}{=} \text{tr}(AP \cdot P^{-1}) = \text{tr}(A)$$

Proof of Characteristic Polynomial/Eigenvalues:

$$\begin{aligned} C_B(\lambda) &= \det(\lambda I - B) = \det(\lambda I - P^{-1}AP) \\ &= \det(\lambda P^{-1}I \cdot P - P^{-1}AP) \\ &= \det(P^{-1}(\lambda I - A)P) \\ &= \det(\lambda I - A) = C_A(\lambda) \end{aligned}$$

Warning:

- Sharing all five properties does NOT guarantee two matrices are similar.

For example: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ & $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

- Being similar is NOT the same as being row equivalent.

Row operations can change determinants, but similarity does not.

Definition: An $n \times n$ matrix A is called **diagonalizable** if $P^{-1}AP$ is a diagonal matrix for some invertible matrix P .

In other words: A is **diagonalizable** if A is *similar* to a diagonal matrix.

Example: Suppose $A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}$. Compute $P^{-1}AP$ using $P = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix}$ (note that $P^{-1} = \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix}$). What does this tell you about A ?

$$P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}.$$

$\Rightarrow A$ is diagonalizable.

What are the eigenvalues of A ?

A & $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ have the same eigenvalues.

What is A^5 ?

$$\begin{aligned} A^5 &= P \cdot \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^5 P^{-1} \\ &= P \cdot \begin{bmatrix} 1 & 0 \\ 0 & 32 \end{bmatrix} \cdot P^{-1} \end{aligned}$$

So diagonalizing a matrix is *nice*. How do we do it?

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ for some unknown λ_i s. Write $P = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n]$. Note that $P^{-1}AP = D$ if and only if $AP = PD$.

What happens when I take PD ?

$$PD = [\vec{v}_1, \dots, \vec{v}_n] \cdot \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

$$= [\lambda_1 \vec{v}_1, \dots, \lambda_n \vec{v}_n]$$

What happens when I take AP ?

$$A \cdot P = A \cdot [\vec{v}_1, \dots, \vec{v}_n]$$

$$= [A\vec{v}_1, \dots, A\vec{v}_n]$$

Theorem: An $n \times n$ matrix A is diagonalizable if and only if A has n eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that the matrix $P = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n]$ is invertible. In this case, $P^{-1}AP = D$, where D is a diagonal matrix such that the elements of D on the main diagonal are the eigenvalues of A .

Said another way:

Theorem: Let A be an $n \times n$ matrix. A is diagonalizable if and only if $\mathbb{R}^n$ has a basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ consisting of eigenvectors of A .

Example: Let $A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$. Is A diagonalizable?

$\lambda_1 = 3, \lambda_2 = -4$ are the two eigenvalues of A .

$$\text{For } \lambda_1 = 3, (3I - A)\vec{x} = \vec{0} \Rightarrow \vec{x} = s \cdot \begin{bmatrix} -7 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda_2 = -4, (-4I - A) \cdot \vec{x} = \vec{0} \Rightarrow \vec{x} = t \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow P = \begin{bmatrix} -7 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow A \text{ is diagonalizable.}$$

Theorem: Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ of an $n \times n$ matrix A . Then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a linearly independent set.

Proof: Prove by contradiction, assume they are L.D.,
WLOG, assume $\vec{x}_1$ can be written as a linear comb.
of the other $k-1$ vectors. Choose a L.I. subset

$$\{\mathbf{x}_{j_1}, \mathbf{x}_{j_2}, \dots, \mathbf{x}_{j_r}\} \subseteq \{\mathbf{x}_1, \dots, \mathbf{x}_k\}, \text{ s.t.}$$

$$\text{Span}\{\mathbf{x}_{j_1}, \dots, \mathbf{x}_{j_r}\} = \text{Span}\{\mathbf{x}_2, \dots, \mathbf{x}_k\}.$$

$$\Rightarrow \mathbf{x}_1 = a_1 \mathbf{x}_{j_1} + a_2 \mathbf{x}_{j_2} + \dots + a_r \mathbf{x}_{j_r}$$

$$\Rightarrow \underline{A\mathbf{x}_1} = A(\underbrace{\mathbf{x}_{j_1} \dots \mathbf{x}_{j_r}}_{\substack{\text{L.I.} \\ \text{set}}}) = a_1 \lambda_{j_1} \mathbf{x}_{j_1} + \dots + a_r \lambda_{j_r} \mathbf{x}_{j_r}$$

Theorem: If A is an $n \times n$ matrix with n distinct eigenvalues, then A is diagonalizable.

Proof:

$$\lambda_1 \mathbf{x}_1 = a_1 \lambda_1 \mathbf{x}_{j_1} + \dots + a_r \lambda_1 \mathbf{x}_{j_r}$$

$$\Rightarrow \underbrace{a_1(\lambda_1 - \lambda_{j_1})}_{\neq 0} \cdot \mathbf{x}_{j_1} + \dots + \underbrace{a_r(\lambda_1 - \lambda_{j_r})}_{\neq 0} \cdot \mathbf{x}_{j_r} = \vec{0}$$

$$\Rightarrow a_1 = a_2 = \dots = a_r = 0.$$

Example: Let $A = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$. Is A diagonalizable?

NO. $(4I - A) \cdot \vec{x} = \vec{0} \Rightarrow \vec{x} = s \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$\Rightarrow E_4(A) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

To get P , we need two L.I. eigen-vectors.

Does an $n \times n$ matrix need to have n distinct eigenvalues to be diagonalizable?

NO, we only need n linearly indep.

eigen-vectors. E.X. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has only

1 eigen-value, but it is diagonalizable.

If the roots of the characteristic polynomial of A are not all distinct, then A may or may not be diagonalizable.

The characteristic polynomial of an $n \times n$ matrix is a degree n polynomial. It always has n **complex** (not necessarily distinct) **roots**, and can be written as the product of n factors:

$$p(\lambda) = (\lambda - \lambda_1)^{k_1} (\lambda - \lambda_2)^{k_2} \dots (\lambda - \lambda_r)^{k_r},$$

where $\lambda_1, \lambda_2, \dots, \lambda_r$ are distinct eigenvalues of A , and $k_1, k_2, \dots, k_r$ are positive integers whose sum is n .

Definition: If A is an $n \times n$ matrix whose characteristic polynomial p is

$$p(\lambda) = (\lambda - \lambda_1)^{k_1} (\lambda - \lambda_2)^{k_2} \dots (\lambda - \lambda_r)^{k_r},$$

then k_i is called the **algebraic multiplicity** of λ_i .

Example: What are the eigenvalues of A and the algebraic multiplicity of each eigenvalue if

$$A = \begin{bmatrix} 2 & 4 & -1 & 3 & 7 \\ 0 & -5 & 2 & 2 & 0 \\ 0 & 0 & 2 & 8 & -3 \\ 0 & 0 & 0 & 3 & 9 \\ 0 & 0 & 0 & 0 & -5 \end{bmatrix} ?$$

$\lambda_3 = 3$, w/ alg. mult. 1.

$\lambda_1 = 2$, w/ alg. mult. 2.
 $\lambda_2 = -5$, w/ alg. mult. 2.

Definition: If λ is an eigenvalue of an $n \times n$ matrix A , the **eigenspace** of A corresponding to λ is

$$E_\lambda(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda\mathbf{x}\}$$

We saw earlier that $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$, so it must have a dimension. In fact,

$$\dim E_\lambda(A) = \dim \text{null}(\lambda I_n - A)$$

$\dim E_\lambda(A)$ is called the **geometric multiplicity** of λ .

Fact: Let λ be an eigenvalue of algebraic multiplicity m of a square matrix A . Then $\dim E_\lambda(A) \leq m$, i.e.

Theorem: A matrix A is diagonalizable if and only if each eigenvalue has geometric multiplicity equal to its algebraic multiplicity.

Example: Find the algebraic and geometric multiplicity of each eigenvalue of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}.$$

For $\lambda_1 = 2$, its alg. mult.
is 2

For $\lambda_2 = 3$, its alg. mult.
is 1.

$$(2I - A)\vec{x} = \vec{0} \Rightarrow \vec{x} = s \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \dim(E_2(A)) = 1.$$

$$(3I - A) \cdot \vec{x} = \vec{0} \Rightarrow \vec{x} = t \cdot \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

$$\Rightarrow \dim(E_3(A)) = 1.$$

So both eigen-values have geo. mult. 1.

Note that, if A is similar to a diagonal matrix D , and the characteristic polynomial of A has *non-real* roots, then the diagonal of D will have *non-real* numbers!

So A is diagonalizable, but not as a *real* diagonal matrix.

Example: Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ be a 2×2 matrix. Is A diagonalizable?

Yes, but not as a real diagonal matrix.

Fact: A symmetric $n \times n$ matrix will have ONLY real eigenvalues.

(Extra examples if time):

Example: Is $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ diagonalizable?

No, since $\underbrace{\text{geo mult.}}_{=2} < \underbrace{\text{alg. mult.}}_{=3}$.

Example: Is $B = \begin{bmatrix} -1 & -3 & 0 \\ 6 & 8 & 0 \\ 3 & 3 & 2 \end{bmatrix}$ diagonalizable?

Yes. For $\lambda_1 = 2$, $\text{geo. mult.} = \text{alg. mult.} = 2$

For $\lambda_2 = 5$, $\text{geo. mult.} = \text{alg. mult.} = 1$.